

DBMS ASSIGNMENT (SY_A , Batch : AS1, Roll_No : 28)

1. Insert 5 employee documents :

```
employees> db.employees.insertMany([
|   { emp_id: 1, name: "Sam", salary: 40000, department: "HR", age: 30 },
|   { emp_id: 2, name: "John", salary: 25000, department: "IT", age: 28 },
|   { emp_id: 3, name: "Sara", salary: 32000, department: "Finance", age: 35 },
|   { emp_id: 4, name: "Steve", salary: 28000, department: "HR", age: 42 },
|   { emp_id: 5, name: "Paul", salary: 50000, department: "IT", age: 33 }
| ])
| {
|   acknowledged: true,
|   insertedIds: {
|     '0': ObjectId('69f33f8169672f0a6b3682d1'),
|     '1': ObjectId('69f33f8169672f0a6b3682d2'),
|     '2': ObjectId('69f33f8169672f0a6b3682d3'),
|     '3': ObjectId('69f33f8169672f0a6b3682d4'),
|     '4': ObjectId('69f33f8169672f0a6b3682d5')
|   }
| }
| }
```

2. Employees in HR with salary>30000 :

```
employees> db.employees.find({
|   department: "HR",
|   salary: { $gt: 30000 }
| })
| [
|   {
|     _id: ObjectId('69f33f8169672f0a6b3682d1'),
|     emp_id: 1,
|     name: 'Sam',
|     salary: 40000,
|     department: 'HR',
|     age: 30
|   }
| ]
```

3. Employees in IT or Finance :

```
employees> db.employees.find({
|   department: { $in: ["IT", "Finance"] }
| })
[
  {
    _id: ObjectId('69f33f8169672f0a6b3682d2'),
    emp_id: 2,
    name: 'John',
    salary: 25000,
    department: 'IT',
    age: 28
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d3'),
    emp_id: 3,
    name: 'Sara',
    salary: 32000,
    department: 'Finance',
    age: 35
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d5'),
    emp_id: 5,
    name: 'Paul',
    salary: 50000,
    department: 'IT',
    age: 33
  }
]
```

4. Employees with age<=40 :

```
employees> db.employees.find({
|   age: { $lte: 40 }
| })
[
  {
    _id: ObjectId('69f33f8169672f0a6b3682d1'),
    emp_id: 1,
    name: 'Sam',
    salary: 40000,
    department: 'HR',
    age: 30
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d2'),
    emp_id: 2,
    name: 'John',
    salary: 25000,
    department: 'IT',
    age: 28
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d3'),
    emp_id: 3,
    name: 'Sara',
    salary: 32000,
    department: 'Finance',
    age: 35
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d5'),
    emp_id: 5,
    name: 'Paul',
    salary: 50000,
    department: 'IT',
    age: 33
  }
]
```

5. Salary>25000 AND age<35 :

```
employees> db.employees.find({
|   salary: { $gt: 25000 },
|   age: { $lt: 35 }
| })
[
  {
    _id: ObjectId('69f33f8169672f0a6b3682d1'),
    emp_id: 1,
    name: 'Sam',
    salary: 40000,
    department: 'HR',
    age: 30
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d5'),
    emp_id: 5,
    name: 'Paul',
    salary: 50000,
    department: 'IT',
    age: 33
  }
]
employees> db.employees.find({
|   name: { $regex: "^S" }
| })
[
  {
    _id: ObjectId('69f33f8169672f0a6b3682d1'),
    emp_id: 1,
    name: 'Sam',
    salary: 40000,
    department: 'HR',
    age: 30
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d3'),
    emp_id: 3,
    name: 'Sara',
    salary: 32000,
    department: 'Finance',
    age: 35
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d4'),
    emp_id: 4,
    name: 'Steve',
    salary: 28000,
    department: 'HR',
    age: 42
  }
]
```

6. Name start with "S" :

```
employees> db.employees.find({
|   name: { $regex: "^S" }
| })
[
  {
    _id: ObjectId('69f33f8169672f0a6b3682d1'),
    emp_id: 1,
    name: 'Sam',
    salary: 40000,
    department: 'HR',
    age: 30
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d3'),
    emp_id: 3,
    name: 'Sara',
    salary: 32000,
    department: 'Finance',
    age: 35
  },
  {
    _id: ObjectId('69f33f8169672f0a6b3682d4'),
    emp_id: 4,
    name: 'Steve',
    salary: 28000,
    department: 'HR',
    age: 42
  }
]
```

7. Name contains "pa" :

```
employees> db.employees.find({
|   name: { $regex: "pa", $options: "i" }
| })
[
  {
    _id: ObjectId('69f33f8169672f0a6b3682d5'),
    emp_id: 5,
    name: 'Paul',
    salary: 50000,
    department: 'IT',
    age: 33
  }
]
```

8. Delete employees with salary>50000 :

```
employees> db.employees.deleteMany({
|   salary: { $gt: 50000 }
| })
{ acknowledged: true, deletedCount: 0 }
```

9. Delete employees where department is "Computer" :

```
employees> db.employees.deleteMany({
|   department: "Computer"
| })
{ acknowledged: true, deletedCount: 0 }
```

10. Delete all documents and collection :

```
employees> db.employees.deleteMany({})
{ acknowledged: true, deletedCount: 5 }
employees> db.employees.drop()
true
```